Status	Finished
Started	Wednesday, 21 October 2026, 9:58 PM
Completed	Wednesday, 21 October 2026, 10:01 PM
Duration	2 mins 50 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By starting from the first element
b. By dividing the list length by two ✓
c. By comparing each element sequentially
d. By using a hash function

Question **2**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search works on unsorted lists
b. Binary search can find multiple instances of the target element
c. Binary search does not require dividing the list
d. Binary search has a lower time complexity on large, sorted lists ✓

Question **3**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. n/2
b. log n
c. n ✓
d. n-1

Question **4**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

- a. In the right sublist ✓
- b. In the left sublist
- c. In the middle of the list
- d. At the beginning of the list

Question **5**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

- a. Linear search ✓
- b. Binary search
- c. Jump search
- d. Hashing

Question **6**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

- a. O(log n) ✓
- b. O(1)
- c. O(n log n)
- d. O👎

Question **7**

Correct

Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

- a. Hashing
- b. Binary search
- c. Dynamic search
- d. Linear search ✓

Question **8**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. log n
b. n ✓
c. n/2
d. 1

Question **9**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

a. Quick search
b. Linear search ✓
c. Merge search
d. Bubble search

Question **10**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

a. When the list is large and sorted
b. When the list is large and unsorted
c. When the list is small and unsorted ✓
d. When the list is small and sorted

Question **11**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. The search is considered unsuccessful ✓
b. The last element is returned
c. An error is raised
d. The first element is returned

Question **12**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The search stops
b. The search continues in the left or right sublist ✓
c. The search continues from the beginning
d. The list is sorted

Question **13**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 89 and 99
b. 89 and 94
c. 90 and 94
d. 90 and 99 ✓

Question **14**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

a. Linear search
b. Binary search ✓
c. Sequential search
d. Hash search

Question **15**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search divides the list into halves.
b. Linear search requires the list to be sorted.
c. Linear search can be applied to both sorted and unsorted lists. ✓
d. Linear search is more efficient than binary search on large lists.